

Epistemics and Instrumentality

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I think the two main ways in which we want to improve upon Policy Selection are:

1. Adding some additional mechanisms that ensure a more delicate and self-conscious treatment of when to update "the reasoner" (in our case, "the prior"), instead of always updating it (this should give us, amongst other things, "the prior washing out").
2. Making epistemics and instrumentality more natively intertwined, instead of sharply separating between them. (Maybe 1. is a special case of this)

The below text is about 2.

Our plan for building the updateless $\mathbf{A}$ is simple:

1. We run an LI $\mathbb{Q}$ up to round $\mathbb{Q}_{10^{10}}$.¹
2. We "freeze" its belief state $\mathbb{Q}_{10^{10}}$. And we extrapolate it into an infinite distribution $\mathbb{P}$.
3. We continue running $\mathbb{Q}$ and use $\mathbb{E}_{\mathbb{P}}(U \mid \mathbb{Q}_{n+k} = \mathcal{Q} \rightarrow A_{n+k} = a)$ to decide actions.

This presents Instrumentality problems: Since we're just copying the unsuspecting $\mathbb{Q}_{10^{10}}$ (that only optimizes for epistemics, being "the observer") and placing it in the center of all our policy actuators (becoming "the actor"), some of its beliefs will not have been optimized for instrumentality. For example, it might have high credence on an epistemically irrelevant φ , while in some part of the tree Ω punishes high $\mathbb{P}(\varphi)$.²

¹This can be done in two ways:

- 1a. We feed $\mathbb{Q}$ "generic mathematical observations", and we just wait until it has good opinions about our tree in particular (since we don't know how to "direct the computational resources" of an LI).
- 1b. Already run $\mathbb{Q}$ inside the tree, letting it decide actions for the early rounds, so its observations are already about the tree.

²Actually, since Ω 's LI "only looks at actions" for its Entanglements, it cannot directly do something like "punishing high $\mathbb{P}(\varphi)$ ". In fact, even our earlier version (that knew $\mathbf{A}$'s algorithm) should only look at the likelihood of sentences $\mathbf{A}(q) = a$. The actual problem is with Logical Coins, and more concretely those that $\mathbb{P}$ thinks are correlated with itself (a particular case is when a simulation of $\mathbf{A}$ is part of the Logical Coin). Then, Ω is free to impose whichever constraints. For example, it can just simulate $\mathbf{A}$ together with the current decision tree, and check indeed whether $\mathbb{P}(\varphi)$ is high. Actually, this problem can also creep into LI Entanglements, since from our agent's perspective they are nothing more than a specially complex kind of Logical Coin. Which assumption can we make that ensures Ω is not being too unfair with these coins, other than the generic "not too many of the Logical Coins turn out false"?

We were looking at the Distributive Oracle procedure to avoid this, but there's a more obvious solution: Instead of copying $\mathbb{Q}_{10^{10}}$ directly (as the finite seed for the infinite $\mathbb{P}$), let $\mathbb{Q}_{10^{10}}$ reason about which finite probability distribution should take that place. That is, just set

$$\mathbb{P}(\varphi) := \operatorname{argmax}_p \mathbb{E}_{\mathbb{Q}_{10^{10}}}(U \mid \mathbb{P}(\varphi) = p)$$

(for all φ such that $\mathbb{Q}_{10^{10}}$ has opinions of that shape), and extend to an infinite distribution. This is just "letting $\mathbb{Q}_{10^{10}}$ design the agent / policy that will play the tree", that is, Policy Selection. This solves the obvious kinds of Instrumentality for $\mathbb{P}$. For example, if a Logical Coin in the tree will have a much better outcome for us in worlds where $\mathbb{P}(\varphi) = p$ is true, $\mathbb{Q}_{10^{10}}$ will take that into account when designing the agent, and will trade off this gain against any undesirable consequences of setting the belief to such an arbitrary value.

Now, what else could we ask for than this? After all, if a change (that would have been an instrumental improvement) is not implemented into $\mathbb{P}$, it will be because $\mathbb{Q}_{10^{10}}$ wasn't aware of this improvement. And we certainly can't demand $\mathbb{Q}_{10^{10}}$ to be aware of everything.

Well, what we'd like is a more native integration of epistemics and instrumentality. Let me be more concrete. If we have obtained the "instrumentality" above, it is by arbitrarily setting a separation between the "purely epistemic" $\mathbb{Q}_{10^{10}}$, and the "optimized for instrumentality" $\mathbb{P}$ (that is, adding one meta-layer of delegation). This still means that the "epistemic" part won't be optimized for instrumentality. For example, if a logical coin in the tree depends on $\mathbb{Q}_{10^{10}}(\varphi)$, we'll again have the problem that this credence is not optimized for instrumentality. We can keep adding delegated layers, but ultimately we'd like something that dissolves the separation completely.

So, how can we have early states of $\mathbb{Q}$ already "constrain themselves" instrumentally (since they know the decision tree they will face)? I'm starting to suspect this won't be possible, for the following reasons.

For really early states, we clearly can't demand anything: they don't even understand basic math about instrumental consequences in the decision tree. That's also why we can't just say " $\mathbb{Q}_{n+1}$ is whatever maximizes U according to $\mathbb{Q}_n$ ": early states will be too silly. But this isn't "a quirk of some early states": it's a fundamental property that happens forever. $\mathbb{Q}_{10^{10}}(\varphi)$ will still "be silly" about things that $\mathbb{Q}_{10^{10}+1}(\varphi)$ knows, just like $\mathbb{Q}_3$ is silly about basic math. This would get fixed if "the observer" decides momentary policies instead of epistemic states... and that's exactly Policy Selection.

Here's a more conceptual and vague argument, contra your "beliefs must look like beliefs":

If you want to "ensure that a bit of your cognition looks like beliefs", someone will be able to look in there and exploit the coherence properties of your beliefs. That is, sometimes there are irreconcilable trade-offs enforced by the "background structure" we've imposed. That's why either you accept that's unfair (and don't worry about it), or you embrace full contextual plasticity.